

Original Manuscript ID: Access-2023-31917

05-12-2023

Original Article Title: “Electricity Theft Detection for Smart Homes: Harnessing the Power of Machine Learning with Real and Synthetic Attacks”

To: IEEE Access Editor

Re: Response to reviewers

Dear Editor,

Thank you for allowing a resubmission of our manuscript, with an opportunity to address the reviewers' comments.

We are uploading (a) our point-by-point response to the comments (below) (response to reviewers, under “Author’s Response Files”), (b) an updated manuscript with yellow highlighting indicating changes (as “Highlighted PDF”), and (c) a clean updated manuscript without highlights (“Main Manuscript”).

Best regards,

Abraham et al.

Reviewer#1, Concern # 1 (please list here): This paper tackles electricity theft detection in smart homes by machine learning with real and synthetic attacks. Overall, this paper is well written. I recommend it for publication. Accept (minor edits)

Author response: We would like to express our sincere gratitude for your positive evaluation of our manuscript. Your acceptance of and suggestions for minor edits are greatly appreciated.

In response to your recommendations for minor edits, I have carefully reviewed the manuscript. During this review, we focused on the following areas to ensure further improvement and alignment with journal standards:

1. Restructuring the paper to enhance consistency.
2. Clarify the proposed framework model and feature selection process.
3. Addressing discrepancies between confusion matrices and Receiver Operating Characteristic (ROC) curves.
4. Expanding the comparative analysis of our model with the existing literature.
5. We have corrected typographical errors, refined captions, labeled tables appropriately, and conducted comprehensive proofreading of the manuscript.

Although specific suggestions or areas of focus were not provided, we endeavored to proactively identify and address potential improvements. We remain open to any specific feedback you or the editorial team may have in the future and are committed to making the necessary adjustments to enhance our manuscript.

Thank you again for your valuable feedback and support.

Author action: We updated the manuscript by

Reviewer#2, Concern # 1 (please list here): The contribution is very weak as thousands of relevant works are doing exactly the same using machine learning in electricity theft detection and evaluated benign and malicious samples.

Author response: We appreciate your insights and the opportunity to further discuss the contribution of our work in the context of the existing literature on machine learning in electricity theft detection in Section I, subsection C – contribution of our proposed models

Author action: We have updated the manuscript by rearranging the introduction and highlighting the contributions of our model to electricity theft detection in smart homes.

Reviewer#2, Concern # 2 (please list here): Are the references provided applicable and sufficient? No

Author response: We have checked and added references as we updated the manuscript

Author action: We updated the manuscript by

Reviewer#3, Concern # 1 (please list here): 1. The main motivation of the paper as well as a summary of the contribution should be stated in the abstract.

Author response: Thank you for taking the time to read and review the manuscript. We appreciate your valuable suggestions and have revised our manuscript accordingly. Below are the responses to your questions and concerns:

Author action: We have updated the manuscript by reformatting the abstract. The introduction section has updated to include the research gap, literature review, main motivation of the paper as well as a summary of the contribution.

Reviewer#3, Concern # 2 (please list here): 2. It is better to divide the Introduction into the following sections.

- a. The motivation for this study is as follows:
- b. Literature Review:
- c. Study gaps:
- d. Contribution:

Author response: Introduction section has been updated accordingly.

Author action: We updated the manuscript by creating Motivation, Literature review, study gaps, and contribution subsections under the Introduction section.

Reviewer#3, Concern # 3 (please list here): 3. The structure of the paper as a whole is inconsistent with the overall review:

Author response: For consistency, we have restructured the paper according to the suggestions above.

Author action: We have updated the manuscript by rearranging each section in our manuscript for consistency.

Reviewer#4, Concern # 1 (please list here):

1) What is the rationale behind classifying the attacks to 5 categories shown in Fig. 1. In other words, given that the proposed framework in Fig. 3 is able to recognize the attack and also the class of the attack, what the importance of such a detection/classification will be for the operator?

Author response: Thank you for taking the time to read and review the manuscript. We appreciate your valuable suggestions and have revised our manuscript accordingly. Below are the responses to your questions and concerns:

Comparing the details of multiclass classification in Figure 1 with the preprocessing for binary classification in Figure 2, we discussed several points regarding their implications within the context of an anomaly detection framework such as the one shown in Figure 3. We updated this in section III of our manuscript.

The importance of the detection/classification was highlighted in section IV of our manuscript

Author action: We updated the manuscript by correcting errors in Fig. 3 caption and also included the benefit of recognizing both specific attacks and unclassified anomalies.

Reviewer#4, Concern # 2 (please list here): 2) Please provide more insights on the model selection and feature selection for the proposed machine learning framework. I recommend including more information about maximum-likelihood parameter estimation while optimizing the objective function.

Author response: We address the model selection in section VI – Experiments results and performance overview

Feature Selection for the Proposed ML Framework was addressed in section V – ETD experiments

Author action: We have updated the manuscript by elaborating on the model and feature selection of our proposed ML framework with relevant references.

Reviewer#4, Concern # 3 (please list here): 3) Please fill the gap between confusion matrices and receiver operating characteristic curves. It is not clear how the curves represent a general view on the content of the confusion matrices. Please kindly explain.

Author response: This is also address in section VI of our updated manuscript – Experiments result and performance evaluation

Author action: We updated the manuscript by updating the figures and table to validate reviewers' comments in section VI of our manuscript.

Reviewer#4, Concern # 4 (please list here): 4) Please report the training and testing errors of the proposed framework. How to guarantee a trade-off between these two errors?

Author response: Our model train and test errors in Section VI, subsection B

Author action: We updated the manuscript by adding more figures and references.

Reviewer#4, Concern # 5 (please list here): 5) The performance of the proposed framework must be compared with other machine learning-based techniques (e.g., SVM) available in the existing literature. Please also report the limitations of the proposed framework.

Author response: We address the comparison with benchmark models in section VII and the limitation in section VIII.

Author action: We updated the manuscript by removing our model limitations from the Discussion and Future Work section and created a subsection for our model limitations.

Reviewer#4, Concern # 6 (please list here): 6) There are several typos throughout the draft (e.g., Table ??, etc.). Please do a second, more focused, proofreading before final decision can be made.

Author response: Thank you for your comments, we reviewed and revised our paper/manuscript thoroughly. We have proofread and corrected the typographical errors throughout the manuscript.

Author action: We updated the manuscript by checking with PaperPal Preflight to check for grammatical issues.

Reviewer#4, Concern # 7 (please list here): 7) Is the subject matter presented in a comprehensive manner? No. It is not.

Author response: We have updated all comments from the reviewers and have represented our manuscript according to the IEEE Access Journal guidelines.

Author action: We updated the manuscript by reorganizing our figures, tables to match relevant sections and IEEE Access Journal guidelines.

Note: *References suggested by reviewers should only be added if it is relevant to the article and makes it more complete. Excessive cases of recommending non-relevant articles should be reported to ieeeaccess@ieee.org*